TRIBHUVAN UNIVERSITY
PATAN MULTIPLE CAMPUS

A CASE STUDY

ON

E-Readiness Index of Nepal: Trends, Strengths and Challenges

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ABSTRACT

E-readiness is the aptitude to leverage the use of information and communication technology and participate in electronic activity such as e-governance. E-readiness of any nation is reflected by the preparedness of data, legal, human, technological and institutional infrastructures. These ICT's core infrastructures are assessed using various metrics and indices including E-participation Index (EPI), Human Capital Index (HCI), Online Service Index (OSI) and Tele-communication Infrastructure Index (TII), that are collectively measured by Electronic Government Development Index (EGDI). The data (2003-2024) shows the progress of Nepal in E-readiness within two decades along with the present status. Further Network Readiness Index (NRI) governed by people, governance, technology and impact provides the insights of both strengths and weaknesses in E-readiness.

Key Words: E-readiness, EPI, HCI, OSI, TII, EGDI, NRI, FTTH, GIOMS

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1. INTRODUCTION

1.1. **OVERVIEW**

This case study report explores the current status of Nepal's E-readiness Index. It evaluates the country's preparedness to adopt Information and Communication Technology (ICT) to digitize the governmental processes. E-readiness refers to the country's ability to leverage ICT for sustainable development, good governance and economic growth. This report assesses key indicators such as human capital, internet penetration, ICT infrastructure and government policies to analyze Nepal's strengths and challenges in embracing digital transformation.

The study incorporates both global benchmarks and national data sourced from the Ministry of Communication and Information Technology (MoCIT) of Nepal to present a comprehensive assessment of the country's digital landscape. This report highlights the nation's progress in areas like e-government and connectivity over the years.

Nepal has steadily improved in digital governance from 2003 to 2024. Its E-Government Development Index value more than doubled, reaching its highest point in 2024 at 0.578. The country's global rank also improved from 132 in 2020 to 119 in 2024, showing progress in online services and digital infrastructure. Nepal is improving its digital services with growing internet use and better access to broadband. FTTH (Fiber to the Home) is available in 74 districts, and 25,000 kilometers of optical fiber have been laid. The GIOMS system is successfully implemented in 41 districts. The Network Readiness Index rank is 109 as per the NRI report 2024.

Despite this progress, Nepal is still lagging behind the global average in e-government development. With an EGDI score of 0.5781 compared to the world average of 0.6382, there remains a gap that highlights the need for continued investment and innovation in digital infrastructure and governance to reach global standards.

1.2. **OBJECTIVES**

1. To assess Nepal's current status in global digital readiness, including online service, e-participation, ICT infrastructure and human capital.
2. To analyze Nepal's performance in the United Nations E-Government Development Index (EGDI) and the Network Readiness Index (NRI).
3. To analyze the national data and reports from the Ministry of Communication and Information Technology (MoCIT) of Nepal.
4. To identify the strengths and challenges in Nepal's journey towards digital transformation.
5. To provide actionable insights and recommendations to improve Nepal's global ranking in e-readiness indicators.

2. **ANALYSIS**

2.1. **THEORETICAL IMPLICATIONS**

E-Government Development Index (EGDI)
EGDI is a composite measure of three dimensions of e-government, namely: provision of online services, telecommunication connectivity and human capacity. It is based on the comprehensive survey of online presence of all 193 United Nations Member States, which assesses national websites and how e-government policies and strategies are applied in general and in specific sectors for delivery of essential services to the public.

Mathematically, the EGDI is a weighted average of three normalized scores on three most important dimensions of e-government, namely: (1) scope and quality of online services (Online Service Index, OSI), (2) development status of telecommunication infrastructure (Telecommunication Infrastructure Index, TII), and (3) inherent human capital (Human Capital Index, HCI). Each of these indices is a composite measure that can be extracted and analysed independently.

$$EGDI = \tfrac{1}{3} (OSI_{normalized} + TII_{normalized} + HCI_{normalized})$$

E-participation Index(EPI) a multifaceted framework, composed of three core components, i.e., e-information, e-consultation and e-decision-making.

Network Readiness Index (NRI)
The Networked Readiness Index is an index published annually by the World Economic Forum in collaboration with INSEAD, as part of their annual Global Information Technology Report. It aims to measure the degree of readiness of countries to exploit opportunities offered by information and communications technology. The four pillars of NRI are technology, people, governance and impact.

2.2. **INTERPRETATION AND EVALUATION**

2.2.1. **Evaluation of Data from Ministry of Communication and Information Technology**

The table below TABLE 1. Shows the data of Nepal's progress in ICT from the study carried out by the Ministry of Communication and Information Technology. This data consists of all the progress of the ministry for digital transformation within a year.

TABLE 1. Achievement up to Paush 2080

S.N.	INDICATORS	FISCAL YEAR 2079/80	UP TO PAUSH 2080
1.	Internet Subscribers Density(%)	130.2	134.76

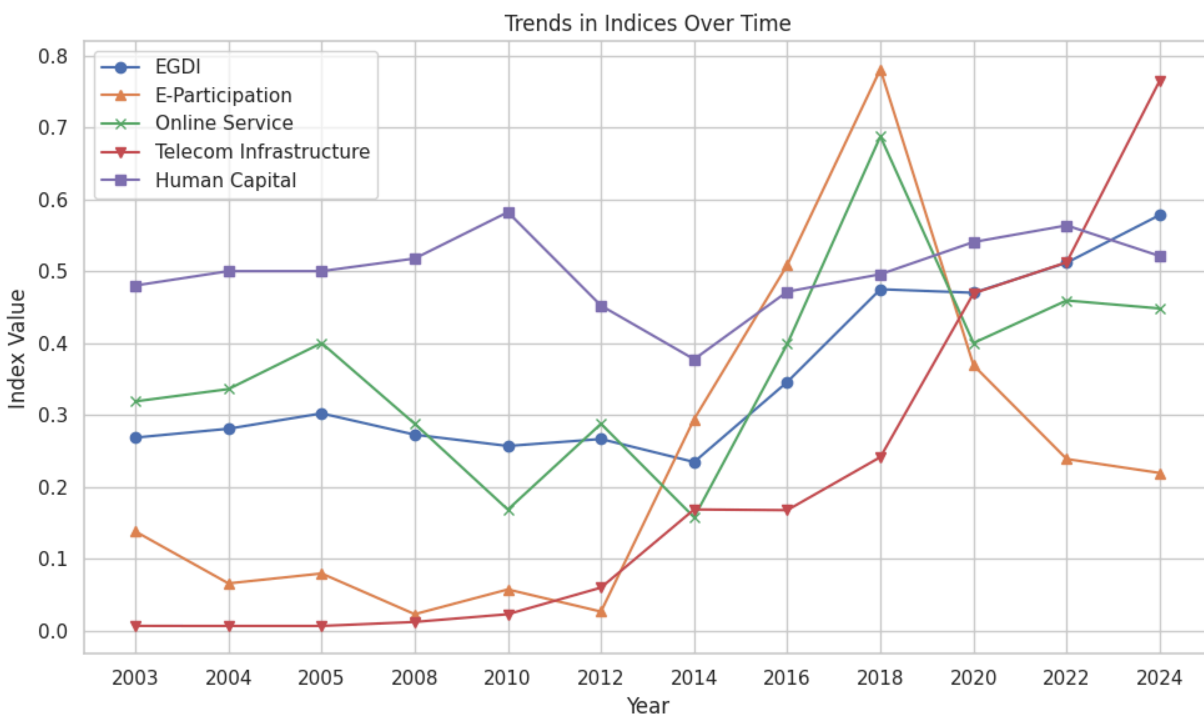
2.	FTTH Service(No. Of districts)	73	74
3.	GIOMS Implementation(No. Of gov-offices)	27	41
4.	Broadband Service(%)	93	94.88
5.	Optical Fiber (Km)	12224	25000
6.	Nagarik App's Services	45	62

Source: [Ministry of Communication and Information Technology]

As of mid-fiscal year 2080, Nepal has continued to expand its digital services and infrastructure. Internet subscriber density has risen from 130.2% in 2079/80 to 134.76%, indicating widespread internet access. FTTH (Fiber to the Home) services are now available in 74 districts, up from 73. The GIOMS system has been implemented in 41 government offices, compared to 27 previously, enhancing administrative efficiency. Broadband coverage has also improved, reaching 94.88% of the population. Notably, the length of optical fiber laid has more than doubled from 12,224 km to 25,000 km, significantly boosting connectivity. Additionally, the Nagarik App now offers 62 public services, up from 45, making digital access to government services more convenient for citizens.

2.2.2. Evaluation of E-government indices

GRAPH 1. E-government Indices of Nepal



Source: [UN E-Government Knowledgebase]

Nepal has made steady progress in digital governance over the years. The E-Government Development Index (EGDI) value has improved from 0.26839 in 2003 to 0.57813 in 2024, showing better digital service delivery and infrastructure. However, Nepal's global rank in e-government is still low at 119, and its score is below the world average of 0.6382, indicating there's still work to be done. The E-Participation Index, which reflects how citizens engage digitally with the government, saw a big drop—from a peak of 0.7809 in 2018 to 0.2192 in 2024, suggesting reduced citizen involvement. Meanwhile, the Online Service Index and Telecommunication Infrastructure Index have both improved significantly, with the latter jumping from just 0.00639 in 2003 to 0.76529 in 2024, pointing to better internet access. The Human Capital Index, which measures education and skills, has remained fairly stable, currently at 0.521. In summary, Nepal is advancing in digital infrastructure and services, but challenges remain in citizen participation and overall global competitiveness.

2.2.3. Evaluation of Network Readiness Index

TABLE 2. Network Readiness Index Data

Country	Score	Rank	Technology	People	Governance	Impact
Nepal	35.96	109	26.57	28.46	39.22	49.59

Source: Network Readiness Index

Nepal ranks 109th out of 133 economies in the NRI index. Its main strength relates to the Impact and the greatest scope for improvement, concerns Governance.

TABLE 3. Nepal's Ranking by Subpillars

Sub-Pillar	Rank	Sub-Pillar	Rank
Content	73	Businesses	108
Quality of Life	80	Inclusion	109
Economy	88	Individuals	110
SDG Contribution	91	Trust	111
Governments	103	Regulations	114
Future Technologies	107	Access	118

Source: Network Readiness Index

When it comes to sub-pillars, the strongest showings of Nepal relate to Content, Quality of Life and Economy. More could be done to improve the economy's performance in trust, regulations and access.

2.3. **WEAKNESSES AND STRENGTHS**

TABLE 4. Highlights of strengths and opportunities for Nepal

Strongest Indicators	Rank	Weakest Indicators	Rank
E-commerce Legislation	1	Data Capabilities	86
Income inequality	22	Adult Literacy Rate	92
AI Scientific Publications	38	Regulation of emerging technologies	111
Mobile App Development	54	Online access to financial account	117
FTTH/building Internet subscriptions	61	E-Participation	117
Rural gap in use of digital payments	66	Handset prices	119
ICT services exports	71	Computer software spending	120
Firms with website	73	Mobile broadband internet traffic within the country	124
Domestic market scale	80	ICT regulatory environment	127
Mobile tariffs	84	Population covered by at least a 3G mobile network	130

Source: [Network Readiness Index]

The indicators where Nepal performed particularly well are E-commerce Legislation and income inequality and AI scientific publications. By Contrast, the country's weakest indicators include population covered by at least a 3G mobile network, ICT regulatory government etc.

2.4. RECOMMENDATIONS

The following are the key recommendations to improve the e-readiness of Nepal:

1. Strengthen Digital Governance and Regulation
    - Upgrade ICT Regulatory Framework: Nepal ranks 127th in ICT regulatory environment and 114th in regulation. Modernize outdated policies, align them with international best practices, and encourage investment-friendly digital laws.
    - Focus on Emerging Technology Regulation: Improve policies around AI, data privacy, and cybersecurity, addressing Nepal's 111th rank in regulation of emerging technologies.

2. Expand Internet Access and Infrastructure
- Reduce the Access Gap: With a low rank of 118 in access and 130th in 3G coverage, Nepal should accelerate rural internet expansion using a mix of FTTH, mobile broadband, and satellite technologies.
- Improve Affordability: Address high handset prices (rank 119) and mobile tariffs (rank 84) through subsidies or partnerships with private telecom providers.

3. Enhance Digital Literacy and Skills
- Focus on Individuals and Inclusion: With Individuals (rank 110) and Inclusion (rank 109) among the weakest areas, implement widespread digital literacy programs, especially for marginalized communities.
- Leverage Adult Literacy: Build upon the moderate adult literacy rate (rank 92) by integrating digital literacy into school and community education.

4. Promote E-Services and Public Engagement
- Expand and Improve Nagarik App: With services growing from 45 to 62, continue enhancing the user experience, add multilingual support, and promote usage awareness.
- Boost E-Participation: Improve Nepal's 117th rank in e-participation by offering more interactive digital platforms for public feedback, digital consultations, and open data initiatives.

5. Encourage Private Sector and Innovation Ecosystem
- Support Local Startups: Improve the Businesses sub-pillar (rank 108) by offering incentives, tax benefits, and incubators for tech startups.
- Increase Firms with Web Presence: With only moderate ranking (73rd) in firms with websites, promote digitization of SMEs through grants and training.

6. Improve Digital Financial Inclusion
- Expand Access to Online Financial Services: With a 117th rank, work with banks and fintech firms to provide simple, mobile-based financial tools for rural populations.
- Close the Rural Digital Payment Gap: Nepal ranks 66th in this area—continue rolling out agent banking and mobile wallet services in remote areas.

3. **CONCLUSION**

Nepal has made noticeable progress in its digital transformation journey, as reflected by improvements in internet penetration, broadband access, and the expansion of e-government services like the Nagarik App. However, the country still faces significant challenges in terms of digital governance, accessibility, regulatory frameworks, and individual digital inclusion, as highlighted by its low ranking in the Network Readiness Index (109th out of 133 countries).

While Nepal shows strong potential in areas such as e-commerce legislation, mobile app development, and AI research, issues like high handset prices, weak regulatory environment, limited access to online financial services, and digital literacy gaps hinder broader digital development.

To advance its e-readiness and effectively transition into a digital economy, Nepal must adopt a multi-dimensional strategy—strengthening policy frameworks, expanding rural connectivity, promoting digital skills, and encouraging innovation. With the right investments, stakeholder collaboration, and inclusive digital policies, Nepal can improve its global standing and ensure digital access and opportunities for all its citizens.

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APPENDICES

Appendix 1: Abbreviations

EGDI – Electronic Government Development Index
EPI – E-Participation Index
HCI – Human Capital Index
OSI – Online Service Index
TII – Telecommunication Infrastructure Index
NRI – Network Readiness Index
FTTH – Fiber to the Home
GIOMS – Government Integrated Office Management System
SME – Small and Medium Enterprises